Haobo Zhao

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Education

University of Virginia, Charlottesville, VA — Sep. 2025 - Present

- *Doctorate in Mechanical Engineering*
- Advisors: Dr. Haibo Dong
- **Research Interests:** Computational Fluid Dynamics, Immersed Boundary Methods, Fluid-Structure Interaction

Johns Hopkins University, Baltimore, MD — Sep. 2023 - May. 2025

- *Master in Mechanical Engineering* — GPA: 3.86/4.0
- Advisors: Dr. Rajat Mittal and Dr. Jung-Hee Seo
- **Coursework:** CFD, Fluid Dynamics I & II, Convection, Numerical Method, Turbulence, Multiphase Flow

Southern Illinois University, Carbondale, IL — 2022 - 2023

- *B.S. in Aviation Technologies* (Dual Degree with SAU) — SIU GPA: 4.0/4.0
- Graduated Magna cum Laude, Dean's List: Fall 2022, Fall 2023

Shenyang Aerospace University, Shenyang, China — 2019 - 2023

- *Bachelor in Aircraft Propulsion Engineering* — GPA: 3.8/4.0
- National Scholarship (2021, top 1% in Department)
- SAU First Class Scholarship (Fall 2019, Fall 2020, Spring 2021)

Research Interests

Computational Fluid Dynamics, Immersed Boundary Methods, Physiological and Biomedical Flows

Publications and Presentations

- Haobo Zhao, Jung-Hee Seo, Venkata Akshintala, Surya Evani, and Rajat Mittal (2026). **Non-invasive Assessment of Pancreatic Duct Hypertension Using Computational Flow Modeling.** *Annals of biomedical engineering*.
- Haobo Zhao, Jung-Hee Seo, Venkata Akshintala, Ibadat Boparai, and Rajat Mittal (2024). **Computational Modeling of Pancreatic Duct Flows for a Novel Non-invasive Diagnosis of Chronic Pancreatitis.** *APS Division of Fluid Dynamics Meeting Abstracts*.

Experience

Johns Hopkins University, Baltimore, MD — Jan. 2024 - Present

- **Master Thesis** (Advisors: Dr. **Rajat Mittal**, Dr. **Jung-Hee Seo) –** Department of Mechanical Engineering
- Developed and validated a patient-specific *CFD* analysis procedure for the pancreatic duct from *MRI* data.
- Collaborated with the JHU medical team led by Dr. Akshintala to compare model predictions with clinical data.
- Formulated and implemented a theoretical flow model to predict pressure variations along the PD.
- Conducted geometry analysis for the theoretical flow model and implemented geometry analysis toolkits, including *VMTK* and *NeuroMorph*.
- Contributed to debugging and enhancing the *NeuroMorph toolkit*.

CFD Visualization Engineer, CertAir LLC — Oct. 2024 - May 2025

- Conducted CFD simulations of cleanroom airflow to optimize layout and design for controlled environments.
- Developed and implemented visualization techniques for airflow patterns and pressure distribution to diagnose cleanroom performance issues.
- Automated processes for robotic pressure measurement, particle detection, and report generation, streamlining

cleanroom certification workflows.
- Collaborated with multidisciplinary teams to enhance cleanroom compliance with ISO standards and improve operational efficiency.

Skills

Programming: Python (automated workflows, plotting, geometry preprocessing), MATLAB, C++, Fortran (computational algorithms)

Software:

- CFD: ANSYS CFX, Fluent, COMSOL
- Geometry Analysis and Visualization: VMTK, NeuroMorph
- (Medical) Geometry Processing: 3D Slicer, Mimics, 3-Matic
- Others: Blender, AutoCAD, SOLIDWORKS

Honors and Awards

- **National Scholarship (2021)**: Awarded to top 1% in department for academic excellence.
- **First Prize, National Mathematics Competition (China, 2020)**: Top 8% of participants.
- **Third Prize, Mechanics Competition of Zhou Peiyuan (China, 2021)**: Recognized for excellence in mechanics.
- **Top 5 in China, iCAN Innovation Contest (2021)**: Ranked 5th out of 3000 teams nationally (Group Award).

Volunteering & Teaching

Student Teacher, Shenyang Aerospace University — Dec. 2019 - Dec. 2021

- Tutored Calculus I & II and College Physics I, including study sessions and final exam reviews.
- Assisted 66 students, improving understanding and academic performance.
- Created final review materials, including mind maps and instructional videos to aid retention.

Core Team Member, SIU-SAU University Collaboration Program — May 2022 - Dec. 2022

- Counseled students from Shenyang Aerospace University in the SIU-SAU Collaboration program.
- Negotiated with SIU's Center for International Education for course arrangements.
- Provided emotional, mental, and financial support in collaboration with professional counselors.

Other Rewards

Mathematics:

- First Prize, National Mathematics Competition for College Students (Top 8%), 2020
- First Prize, Mathematical Modeling Competition in Liaoning Province, 2021
- Second Prize, Mathematical Modeling Competition in Liaoning Province, 2020
- Second Prize, National Mathematics Competition for College Students, 2021
- Third Prize, China CUMCM (Contemporary Undergraduate Mathematical Contest in Modeling), 2021
- Third Prize, MathorCup Undergraduate Mathematical Modeling Challenge, 2021
- Second Prize, Asia Pacific Mathematical Contest in Modeling, 2020
- Second Prize, Mathematical Contest in Modeling of Three Provinces in Northeast China, 2020

Physics:

- Third Prize, Mechanics Competition in Honour of Zhou Peiyuan, 2021
- First Prize, Physics Experiment Competition in Liaoning Province, 2020

Innovation and Design:

- iCAN Innovation Contest (Finalist), Top 5 in China, 2021
- Third Prize, China College Students' "Internet+" Innovation and Entrepreneurship Competition, 2021

- Second Prize, The 6th China International “Internet+” College Student Innovation and Entrepreneurship Competition SAU Selection, 2020
- First Prize, SAU Future Engine Design Competition